

- Trace amounts may be present in nursing children whose mothers are on fluoxetine
- If child becomes irritable or sedated, breast feeding or drug may need to be discontinued
- Immediate postpartum period is a high-risk time for depression, especially in women who have had prior depressive episodes, so drug may need to be reinstituted late in the third trimester or shortly after childbirth to prevent a recurrence during the postpartum period
- Must weigh benefits of breast feeding with risks and benefits of antidepressant treatment versus nontreatment to both the infant and the mother
- For many patients this may mean continuing treatment during breast feeding

The art of Psychopharmacology

Potential Advantages

- Patients with atypical depression (hypersomnia, increased appetite)
- Patients with fatigue and low energy
- Patients with comorbid eating and affective disorders
- Generic is less expensive than brand name where available
- Patients for whom weekly administration is desired
- Children with OCD or depression

Potential Disadvantages

- Patients with anorexia
- Initiating treatment in anxious, agitated patients
- Initiating treatment in severe insomnia

Primary Target Symptoms

- Depressed mood
- Energy, motivation, and interest
- Anxiety (eventually, but can actually increase anxiety, especially short-term)
- Sleep disturbance, both insomnia and hypersomnia (eventually, but may actually cause insomnia, especially short-term)

Pearls

- * May be a first-line choice for atypical depression (e.g., hypersomnia, hyperphagia, low energy, mood reactivity)
- Consider avoiding in agitated insomniacs
- Can cause cognitive and affective “flattening”
- Not as well tolerated as some other SSRIs for panic disorder and other anxiety disorders, especially when dosing is initiated, unless given with co-therapies such as benzodiazepines or trazodone
- Long half-life; even longer-lasting active metabolite
- * Actions at 5HT_{2C} receptors may explain its activating properties

- * Actions at 5HT_{2C} receptors may explain in part fluoxetine's efficacy in combination with olanzapine for bipolar depression and treatment-resistant depression, since both agents have this property
- For sexual dysfunction, can augment with bupropion, sildenafil, vardenafil, or tadalafil, or switch to a non-SSRI such as bupropion or mirtazapine
 - Mood disorders can be associated with eating disorders (especially in adolescent females) and be treated successfully with fluoxetine
 - SSRIs may be less effective in women over 50, especially if they are not taking estrogen
 - SSRIs may be useful for hot flushes in perimenopausal women
 - Some postmenopausal women's depression will respond better to fluoxetine plus estrogen augmentation than to fluoxetine alone
 - Nonresponse to fluoxetine in elderly may require consideration of mild cognitive impairment or Alzheimer disease
 - SSRIs may not cause as many patients to attain remission of depression as some other classes of antidepressants (e.g., SNRIs)
 - A single pill containing both fluoxetine and olanzapine is available for combination treatment of bipolar depression, psychotic depression, and treatment-resistant unipolar depression

Suggested Reading

Anderson IM . Selective serotonin reuptake inhibitors versus tricyclic antidepressants: a meta-analysis of efficacy and tolerability . *J Affect Disord* 2000 ;**58** :19 –36 .

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March JS , Silva S , Petrycki S , et al. The treatment for adolescents with depression study (TADS): long-term effectiveness and safety outcomes . *Arch Gen Psychiatry* 2007 ;**64** (10):1132 –43.

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Flupenthixol

Therapeutics

Brands

- Depixol
- *see index for additional brand names*

Generic?

No

Class

- Neuroscience-based Nomenclature: dopamine receptor antagonist (D-RAn)
- Conventional antipsychotic (neuroleptic, thioxanthene, dopamine 2 antagonist)

Commonly Prescribed for

(bold for FDA approved)

- Schizophrenia
- Depression (low dose)
- Other psychotic disorders
- Bipolar disorder

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis

How Long Until It Works

- With injection, psychotic symptoms can improve within a few days, but it may take 1–2 weeks for notable improvement
- With oral formulation, psychotic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior

If It Works

- Most often reduces positive symptoms in schizophrenia but does not eliminate them
- Most schizophrenic patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
- Continue treatment in schizophrenia until reaching a plateau of improvement
- After reaching a satisfactory plateau, continue treatment for at least a year after first episode of psychosis in schizophrenia
- For second and subsequent episodes of psychosis in schizophrenia, treatment may need to be indefinite
- Reduces symptoms of acute psychotic mania but not proven as a mood stabilizer or as an effective maintenance treatment in bipolar disorder

- After reducing acute psychotic symptoms in mania, switch to a mood stabilizer and/or an atypical antipsychotic for mood stabilization and maintenance

If It Doesn't Work

- Consider trying one of the first-line atypical antipsychotics
- Consider trying another conventional antipsychotic
- If 2 or more antipsychotic monotherapies do not work, consider clozapine

Best Augmenting Combos for Partial Response or Treatment Resistance

- Augmentation of conventional antipsychotics has not been systematically studied
- Addition of a mood-stabilizing anticonvulsant such as valproate, carbamazepine, or lamotrigine may be helpful in both schizophrenia and bipolar mania
- Augmentation with lithium in bipolar mania may be helpful
- Addition of a benzodiazepine, especially short-term for agitation

Tests

- * Since conventional antipsychotics are frequently associated with weight gain, before starting treatment, weigh all patients and determine

if the patient is already overweight (BMI 25.0–29.9) or obese (BMI >30)

- Before giving a drug that can cause weight gain to an overweight or obese patient, consider determining whether the patient already has pre-diabetes (fasting plasma glucose 100–125 mg/dL), diabetes (fasting plasma glucose >126 mg/dL), or dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol), and treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management
- * Monitor weight and BMI during treatment
- * Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics
- * While giving a drug to a patient who has gained >5% of initial weight, consider evaluating for the presence of pre-diabetes, diabetes, or dyslipidemia, or consider switching to a different antipsychotic
- Monitoring elevated prolactin levels of dubious clinical benefit
- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and flupenthixol should be discontinued at the first sign of decline of WBC in the absence of other causative factors

- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side Effects

How Drug Causes Side Effects

- Acute blockade of dopamine 2 receptors in the striatum can cause drug-induced parkinsonism, dystonia, or akathisia
- Chronic blockade of dopamine 2 receptors in the striatum can cause tardive dyskinesia
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin
- By blocking dopamine 2 receptors excessively in the mesocortical and mesolimbic dopamine pathways, especially at high doses, it can cause worsening of negative and cognitive symptoms (neuroleptic-induced deficit syndrome)
- Blocking muscarinic cholinergic receptors can cause dry mouth, blurred vision, urinary retention, constipation, and paralytic ileus
- Antihistaminic actions may cause sedation, weight gain
- Blocking alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope
- Mechanism of weight gain and any possible increased incidence of diabetes or dyslipidemia with conventional antipsychotics is

unknown

Notable Side Effects

- * Neuroleptic-induced deficit syndrome
- * Drug-induced parkinsonism
- * Insomnia, restlessness, agitation, sedation
- * Tardive dyskinesia (risk increases with duration of treatment and with dose)
- * Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration
- * Galactorrhea, amenorrhea
- Tachycardia
- Weight gain
- Hypomania
- Rare eosinophilia



Life-Threatening or Dangerous Side Effects

- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure
- Rare seizures

- Rare jaundice, leukopenia
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis

Weight Gain



- Many experience and/or can be significant in amount

Sedation



- Occurs in significant minority

What to Do About Side Effects

- Wait
- Wait
- Wait
- For drug-induced parkinsonism, add an anticholinergic agent
- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Reduce the dose
- For sedation, give at night
- Switch to an atypical antipsychotic

- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain

Best Augmenting Agents for Side Effects

- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing And Use

Usual Dosage Range

- Oral 3–6 mg/day in divided doses
- Intramuscular 40–120 mg every 1–4 weeks

Dosage Forms

- Tablet 0.5 mg, 3 mg
- Injection 20 mg/mL, 100 mg/mL

How to Dose

- Oral: initial 1 mg 3 times a day; increase by 1 mg every 2–3 days; maximum generally 18 mg/day
- Intramuscular: initial dose 20 mg for patients who have not been exposed to long-acting depot antipsychotics, 40 mg for patients who have previously demonstrated tolerance to long-acting depot antipsychotics; after 4–10 days can give additional 20 mg dose; maximum 200 mg every 1–4 weeks



Dosing Tips

- The peak of action for the decanoate is usually 7–10 days, and doses generally have to be administered every 2–3 weeks
- May have more activating effects at low doses, which can sometimes be useful as a second-line, short-term treatment of depression
- Some evidence that flupenthixol may improve anxiety and depression at low doses
- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly
- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine

- Children and elderly should generally be dosed at the lower end of the dosage spectrum
- Treatment should be suspended if absolute neutrophil count falls below $1000/\text{mm}^3$

Overdose

- Agitation, confusion, sedation, drug-induced parkinsonism, respiratory collapse, circulatory collapse

Long-Term Use

- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- Slow down-titration of oral formulation (over 6–8 weeks), especially when simultaneously beginning a new antipsychotic while switching (i.e., cross-titration)
- Rapid oral discontinuation may lead to rebound psychosis and worsening of symptoms
- If antiparkinson agents are being used, they should be continued for a few weeks after flupenthixol is discontinued

Pharmacokinetics

- Oral: maximum plasma concentrations within 3–8 hours
- Intramuscular: rate-limiting half-life approximately 8 days with single dose, approximately 17 days with multiple doses



Drug Interactions

- May decrease the effects of levodopa, dopamine agonists
- May increase the effects of antihypertensive drugs except for guanethidine, whose antihypertensive actions flupenthixol may antagonize
- CNS effects may be increased if used with other CNS depressants
- Combined use with epinephrine may lower blood pressure
- Ritonavir may increase plasma levels of flupenthixol
- May increase carbamazepine plasma levels
- Some patients taking a neuroleptic and lithium have developed an encephalopathic syndrome similar to neuroleptic malignant syndrome



Other Warnings/Precautions

- If signs of neuroleptic malignant syndrome develop, treatment should be immediately discontinued
- Use cautiously in patients with alcohol withdrawal or convulsive disorders because of possible lowering of seizure threshold

- In epilepsy patients, dose 10–20 mg every 15 days for intramuscular formulation
- Use with caution if at all in patients with Parkinson's disease, severe arteriosclerosis, or Lewy body dementia
- Possible antiemetic effect of flupenthixol may mask signs of other disorders or overdose; suppression of cough reflex may cause asphyxia
- Avoid extreme heat exposure
- Do not use epinephrine in event of overdose as interaction with some pressor agents may lower blood pressure
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy
- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- If patient is taking a large concomitant dose of a sedative hypnotic
- If patient has CNS depression
- If patient is comatose or if there is brain damage

- If there is blood dyscrasia
- If patient has pheochromocytoma
- If patient has liver damage
- If patient has a severe cardiovascular disorder
- If patient has renal insufficiency
- If patient has cerebrovascular insufficiency
- If there is a proven allergy to flupenthixol

Special Populations

Renal Impairment

- Oral: recommended to take half or less of usual adult dose
- Intramuscular: recommended dose schedule generally 10–20 mg every 15 days

Hepatic Impairment

- Use with caution
- Oral: recommended to take half or less of usual adult dose

Cardiac Impairment

- Use with caution
- Oral: recommended to take half or less of usual adult dose

Elderly

- Intramuscular: recommended initial dose generally 5 mg; recommended dose schedule generally 10–20 mg every 15 days
- Oral: recommended to take half or less of usual adult dose
- Although conventional antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with antipsychotics are at increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events



Children and Adolescents

- Not recommended for use in children



Pregnancy

- Not recommended for use during pregnancy
- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding
- Reports of drug-induced parkinsonism, jaundice, hyperreflexia, hyporeflexia in infants whose mothers took a conventional

antipsychotic during pregnancy

- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary
- Atypical antipsychotics may be preferable to conventional antipsychotics or anticonvulsant mood stabilizers if treatment is required during pregnancy

Breast Feeding

- Some drug is found in mother's breast milk
- * Recommended either to discontinue drug or bottle feed

The Art Of Psychopharmacology

Potential Advantages

- Noncompliant patients

Potential Disadvantages

- Children
- Elderly
- Patients with tardive dyskinesia

Primary Target Symptoms

- Positive symptoms of psychosis

- Negative symptoms of psychosis
- Aggressive symptoms

 Pearls

- Less sedation and orthostatic hypotension but more drug-induced parkinsonism than some other conventional antipsychotics
- Patients have very similar antipsychotic responses to any conventional antipsychotic, which is different from atypical antipsychotics where antipsychotic responses of individual patients can occasionally vary greatly from one atypical antipsychotic to another
- Patients with inadequate responses to atypical antipsychotics may benefit from a trial of augmentation with a conventional antipsychotic such as flupenthixol or from switching to a conventional antipsychotic such as flupenthixol
- However, long-term polypharmacy with a combination of a conventional antipsychotic such as flupenthixol with an atypical antipsychotic may combine their side effects without clearly augmenting the efficacy of either
- For treatment-resistant patients, especially those with impulsivity, aggression, violence, and self-harm, long-term polypharmacy with 2 atypical antipsychotics or with 1 atypical antipsychotic and 1 conventional antipsychotic may be useful or even necessary while closely monitoring

- In such cases, it may be beneficial to combine 1 depot antipsychotic with 1 oral antipsychotic
- Although a frequent practice by some prescribers, adding 2 conventional antipsychotics together has little rationale and may reduce tolerability without clearly enhancing efficacy

Suggested Reading

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Fluphenazine

Therapeutics

Brands

- Prolixin
- *see index for additional brand names*

Generic?

Yes

Class

- Conventional antipsychotic (neuroleptic, phenothiazine, dopamine 2 antagonist)

Commonly Prescribed for

(bold for FDA approved)

- **Psychotic disorders**
- Bipolar disorder

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis

How Long Until It Works

- Psychotic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior

If It Works

- Most often reduces positive symptoms in schizophrenia but does not eliminate them
- Most schizophrenic patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
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Tests

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- Before giving a drug that can cause weight gain to an overweight or obese patient, consider determining whether the patient already has pre-diabetes (fasting plasma glucose 100–125 mg/dL), diabetes (fasting plasma glucose >126 mg/dL), or dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL

cholesterol), and treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

- * Monitor weight and BMI during treatment
- * Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics
- * While giving a drug to a patient who has gained >5% of initial weight, consider evaluating for the presence of pre-diabetes, diabetes, or dyslipidemia, or consider switching to a different antipsychotic
- Should check blood pressure in the elderly before starting and for the first few weeks of treatment
- Monitoring elevated prolactin levels of dubious clinical benefit
- Phenothiazines may cause false-positive phenylketonuria results
- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and fluphenazine should be discontinued at the first sign of decline of WBC in the absence of other causative factors
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)